Bring Your Own Agent to Sheets

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Context-aware spreadsheet intelligence with bidirectional agent synchronization

Product thesis: Your AI already knows your work. Your spreadsheet shouldn't make it start over.

1. Executive summary

Bring Your Own Agent to Sheets is an AI-powered Google Sheets integration that connects an existing AI agent and its authorized context to live spreadsheet workflows. Users can analyze data, propose edits, build formulas, generate charts and forecasts, and receive recommendations grounded in their company goals. Changes require user approval. Users can subsequently review and save analysis summaries to a supported agent-context destination.

Vision: Preserve context between AI conversations and the spreadsheets where financial work happens.

- Bring your context: reuse authorized agent instructions, knowledge and relevant conversation context where the provider supports it.
- Turn conversations into actions: create spreadsheet formulas, charts, forecasts and recommendations.
- Maintain control: preview, validate and approve proposed changes.
- Keep context current: explicitly sync approved models, assumptions and insights to a supported destination.

2. Problem definition

Financial analysts move between AI conversations, Google Sheets, company documents and planning tools. Context built up in one tool may not be available in another. Analysts must restate company objectives and assumptions when modeling, then manually transfer the new results back into their AI workflows.

Problem statement: Financial analysts lack a reliable, continuous connection between the AI agents that understand their business context and the spreadsheets where they perform financial analysis.

Research hypotheses

Hypothesis	Validation question
H1 — Context switching is painful	How often do users copy or repeat information between AI and Sheets?
H2 — Context improves relevance	Are company-grounded recommendations more actionable?
H3 — Users want controlled edits	Will users approve proposed agent-driven spreadsheet changes?
H4 — Bidirectional sync adds value	Will users reuse saved results in future conversations?
H5 — Trust drives adoption	Do previews, evidence and undo improve confidence?

3. Target users

Primary persona: FP&A analyst at a growing technology company (illustrative). Uses Google Sheets for revenue, expense, budget and forecast reporting and uses an AI assistant for business questions. Wants accurate, context-aware analysis without repeated explanation.

Primary jobs to be done:

- Build a forecast using already-known company goals and assumptions.
- Review and approve any agent-proposed changes.
- Preserve important assumptions and results for future conversations.
- Inspect the evidence behind recommendations.

Secondary users: finance managers, spreadsheet-heavy startup founders, operations analysts. MVP scope: one analyst, one Sheet, one agent connection.

4. Goals and non-goals

Goals	Non-goals for MVP
Connect one supported agent/context source	Universal provider compatibility
Read and modify Google Sheets	Excel support
Create formulas, forecasts and charts	Full-scale financial modeling engine
Require previews and approvals	Autonomous unrestricted editing
Save analysis to supported context destination	Writing to proprietary memory without permission
Generate grounded recommendations	Guaranteed financial advice

5. End-to-end user journey

1. Connect an existing agent and inspect available context.
2. Add more company context: documents, goals and assumptions.
3. Work in Google Sheets through natural-language requests.
4. Preview proposed formulas, cell edits, charts and model assumptions.
5. Approve and apply selected changes to the Sheet.
6. Receive insights grounded in company context and updated calculations.
7. Explore what-if scenarios without overwriting baseline data.
8. Review and sync an approved summary back to the agent's supported context destination.

6. Functional requirements

FR-01 Connect an existing AI agent

Priority: P0

Expected behavior: Select and authenticate one supported agent/context provider; retrieve only authorized accessible context; display capabilities and sources; allow disconnect.

Acceptance criteria: Connected status is visible; inaccessible history is not implied; failures are actionable; no Sheet writes occur during connection.

FR-02 Add more company context

Priority: P0

Expected behavior: Upload supported documents or add goals and assumptions; distinguish inherited, uploaded and inferred context; enable source selection.

Acceptance criteria: User can add/remove sources; provenance is visible; missing and conflicting sources are surfaced.

FR-03 Read and understand spreadsheet data

Priority: P0

Expected behavior: Read authorized values, formulas, tabs and ranges; detect headers, currencies and dates; ask clarifying questions.

Acceptance criteria: Correctly identifies predefined data columns; cites ranges; never invents missing figures.

FR-04 Create and edit cells and formulas

Priority: P0

Expected behavior: Create columns, fill formulas, update ranges, format values, add forecast rows and charts using constrained operations.

Acceptance criteria: All operations are structured and validated; unauthorized ranges are untouched; errors are reported.

FR-05 Preview and approve changes

Priority: P0

Expected behavior: Show before/after values, formulas, affected ranges and assumptions; support approval, rejection and revision.

Acceptance criteria: No write before approval; source changes invalidate old previews; approved operations match executed operations.

FR-06 Forecasts and visualizations

Priority: P0

Expected behavior: Generate line/bar charts, growth and margin metrics and a transparent basic three-month forecast; distinguish actuals and projections.

Acceptance criteria: Charts use correct ranges; forecast method and assumptions are explicit; calculations are reproducible.

FR-07 Context-aware recommendations

Priority: P0

Expected behavior: Combine deterministic spreadsheet calculations with company objectives; identify supporting data; distinguish findings from suggestions.

Acceptance criteria: Company-specific targets appear only if available in approved context; uncertainty and sources are visible.

FR-08 What-if scenarios

Priority: P1

Expected behavior: Modify assumptions in a separate scenario; compare against baseline; optionally save approved scenario.

Acceptance criteria: Baseline remains unchanged; changed assumptions and impacts are explicit.

FR-09 Sync context back to agent

Priority: P1

Expected behavior: Produce a structured summary of models, assumptions, results and sources; let user select items and destination; require approval.

Acceptance criteria: Destination confirms persistence; hypothetical scenarios are labeled; local storage is never misrepresented as external memory sync.

FR-10 Audit and recovery

Priority: P1

Expected behavior: Record requests, proposed operations, approvals, executions, errors and context updates; support recovery of affected cells where possible.

Acceptance criteria: User can review history and distinguish partial failure from success.

7. Functional scope and prioritization

Capability	Priority
Connect an existing AI agent	P0
Add more company context	P0
Read and understand spreadsheet data	P0
Create and edit cells and formulas	P0
Preview and approve changes	P0
Forecasts and visualizations	P0
Context-aware recommendations	P0
What-if scenarios	P1
Sync context back to agent	P1
Audit and recovery	P1

Hackathon demo stretch: what-if scenario and real context synchronization. Future: multiple providers, organization-wide permissions, advanced modeling, scheduled analysis and collaboration.

8. Information architecture and UX principles

9. Agent setup: connect → review inherited context → add more context → permissions.
10. Sheets workspace: chat → analyze → preview → approve → execute → insights → scenarios.
11. Context management: review summary → choose items → approve sync → verify persistence.

- Make context and its provenance visible.
- Propose before acting; require explicit approval for writes.
- Use deterministic computation for financial arithmetic.
- Ground recommendations in source ranges and company documents.
- Never silently save hypothetical scenarios as established facts.

9. Technical architecture

Recommended hackathon architecture: Next.js/React companion app connected to the Google Sheets API, with a backend orchestrator and a single verified agent/context integration.

Layer	Technology	Purpose
Frontend	Next.js, React, Tailwind	Chat, preview, approvals, results
Spreadsheet	Google Sheets API	Read/write cells and create charts
Auth	Google OAuth 2.0	Scoped authorized access
Backend	FastAPI or Next.js routes	Operation validation and execution
Agent integration	Provider adapter; MCP if supported	Retrieve authorized agent context
Database	PostgreSQL/Supabase	Context, approvals, execution history
Calculation	Python / Sheets formulas	Deterministic verification

Execution pipeline: User request → approved context → Sheets read → structured action plan → validation → preview → approval → source version check → execute → verify → insights → optional approved context sync.

Example operation: write_formula to Sheet1!F2:F7 using =B{row}-C{row}; create a three-month moving-average forecast; create a line chart from a validated range. All operations are allowlisted and checked server-side.

Interoperability limitation: MCP does not itself expose proprietary conversation history or grant permission to update agent memory. Demonstrate one genuinely supported integration; distinguish external sync from local storage.

10. Data model

Entity	Key fields
User	id, preferences
AgentConnection	provider, capabilities, permissions, status
ContextSource	type, provenance, version, reference
Spreadsheet	id, owner, source snapshot
AnalysisSession	request, context sources, sheet reference
OperationPlan	operations, validation, status
Approval	plan, user, decision, timestamp
ExecutionRecord	results, errors, before/after
ForecastModel	method, assumptions, source and output ranges
ContextUpdate	summary, destination, approval, sync status

11. Non-functional requirements

Area	Requirement
Security	Encrypted server-side OAuth tokens; least privilege
Privacy	Authorized access to company financial data only
Reliability	No false success status; partial failures visible
Performance	Design targets: <5s simple read; <15s common planning, excluding approval
Transparency	Inspectable assumptions, edits and sources
Accessibility	Keyboard navigation and WCAG 2.2 AA design targets
Prompt safety	Treat imported documents and spreadsheet cells as untrusted data
Retention	Explicit retention and deletion rules before production

12. Edge cases and error handling

Scenario	Response
Provider cannot expose history	Explain limitation; allow supported import
Context conflicts	Surface conflicting sources
Read-only spreadsheet	Disable edits; explain permissions
Ambiguous headers	Request range selection
Invalid formula	Flag before execution
Sheet changes after preview	Refresh plan and require new approval
Insufficient forecast data	Explain limitations; do not fabricate
Partial write failure	Report completed actions and recovery path
Context destination read-only	Offer export/local save without claiming sync
Hypothetical scenario	Label and exclude from factual sync by default

13. Success metrics

North star: Percentage of completed analysis sessions with an approved spreadsheet action successfully executed using connected context.

First-round measure	Proposed target
Understand core proposition	80%
Complete main flow unassisted	80%
Understand available context	90%
Understand approval boundary	100%
Predefined demo operations succeed	100%
Find context continuity useful	70%

Targets are hypotheses, not observed benchmarks. Long-term metrics include time saved, repeated-context reduction, acceptance/reversal rates, sync success and repeat usage.

14. User research and usability testing

Conduct 5–8 moderated interviews with spreadsheet-heavy financial analysts and FP&A professionals who already use AI assistants.

12. Connect an agent and identify accessible context.
13. Request a profit column and revenue forecast.
14. Review and approve proposed edits.
15. Find the source behind a recommendation.
16. Explore a 10% marketing reduction scenario.
17. Save an approved analysis summary to a supported context store.

Interview prompts: Describe the last time you used AI with a financial spreadsheet. What information did you copy between tools? What would you expect a connected agent to know? Which edits require approval? How would you verify a forecast? Would reusing a saved analysis matter? What would prevent adoption?

15. Competitive positioning

Category	Existing value	Hypothesized differentiation
AI spreadsheet assistants	Analyze/edit Sheets	Bring context from existing agent
General AI assistants	Reason over documents/conversations	Execute approved live Sheets operations
Automation tools	Connect apps and tasks	Conversational, context-aware modeling and sync

Competitors/substitutes include Gemini in Sheets, Microsoft Copilot, spreadsheet AI tools and custom agents. Differentiation is a hypothesis requiring competitive validation.

16. Hackathon implementation plan

Milestone 1 — Connect

Connect a real context source and Google Sheet. Definition of done: answer a question requiring both spreadsheet data and an approved company goal.

Milestone 2 — Act

Implement structured operations, preview, approval, formula insertion and one chart. Definition of done: user approves and sees real changes in Sheets.

Milestone 3 — Close the loop

Generate an analysis summary and persist it to a supported context destination. Definition of done: a later interaction retrieves approved assumptions without restating them.

Role	Ownership
Product/design	Flow, approval UX, testing, demo
Frontend	Chat, connection, preview, results
Backend	Sheets API, operations, validation
AI/integrations	Agent adapter, retrieval, sync

17. Demo narrative

18. Problem: AI understands the company, but Sheets often lacks that context.
19. Connect: show authorized inherited context.
20. Analyze: request a Q4 forecast grounded in company goals.

21. Approve: show proposed formulas, chart and assumptions before writes.
22. Interpret: show grounded business insights.
23. Sync: save an approved summary and retrieve it in a later agent interaction.

Closing: Your agent comes with context, works directly in your spreadsheet, and carries approved insights forward.

18. Risks and dependencies

Risk	Mitigation
External agent interoperability	Verify one real provider integration and disclose limitations
Financial accuracy	Deterministic arithmetic, constrained operations, evidence and approvals
Context contamination	Separate actuals, forecasts, hypotheticals and recommendations
Scope expansion	One provider, one Sheet, one forecast, one chart, one sync destination
Sensitive data	Scoped permissions, token protection, retention controls

19. Open product decisions

24. Agent integration: provider-specific integration, MCP-compatible context source, or manual import first?
25. Initial surface: web companion or native Sheets add-on?
26. Context destination: external agent store, product-managed store, or export?
27. Which financial model and test dataset are the first supported scenario?
28. What audit, recovery and privacy guarantees are necessary before real-company pilots?

20. Final product definition

Target: FP&A professionals using Google Sheets and AI for forecasting and analysis. Problem: disconnected context across chat and spreadsheets. Solution: bring authorized agent context into Sheets, execute approved actions, generate grounded recommendations and sync approved results back to a supported destination.

MVP success: Connect one real context source, analyze a real Sheet, approve a formula and chart, receive a grounded recommendation and reuse an approved analysis summary in a subsequent interaction.

Product differentiator: Context continuity, not merely AI spreadsheet editing.